Decision Analytics Journal 9 (2023) 100320

Contents lists available at ScienceDirect

Decision Analytics Journal

journal homepage: www.elsevier.com/locate/dajour

A logic Petri net model for dynamic multi-agent game decision-making

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ARTICLE INFO

Keywords:
Logic Petri nets
Multi-agent decision problems
Dynamic game decision-making
Optimization
Emergency decision-making
Resource conflicts

ABSTRACT

This study proposes a logical Petri net model to leverage the modeling advantages of Petri nets in handling batch processing and uncertainty in value passing and to integrate relevant game elements from multi-agent game processes for modeling multi-agent decision problems and resolving optimization issues in dynamic multi-agent game decision-making. Firstly, the attributes of each token are defined as rational agents, and utility function values and state probability transition functions are assigned to them. Secondly, decision transitions are introduced, and the triggering of the optimal decision transition is determined based on a comparison of token utility function values, along with an associated algorithm. Finally, a dynamic game emergency business decision-making process for sudden events is modeled and analyzed using the logic game decision Petri net. Based on reachable markings, reachable graphs are constructed to analyze the dynamic game process. Algorithms are described for the generation of reachable graphs, and the paper explores how the logic game decision model for sudden events can address dynamic game decision problems, generate optimal emergency plans, and analyze resource conflicts during emergency processes. The effectiveness and superiority of the model in analyzing the emergency business decision-making process for sudden events are validated.

1. Introduction

A sudden event is an emergency that poses direct risks and impacts human health, life, and property, requiring urgent intervention to prevent further deterioration. These intervention measures are organized into a process [1], which is typically described in an emergency plan and referred to as the emergency response process. In this process, all emergency personnel are dedicated to managing disasters to minimize or avoid the secondary impacts of the disaster. Generating better contingency plans before emergency responses have become an urgent issue to address. Ref. [2] studied the uncertainty of evacuation time during emergencies, and its stochastic analysis was conducted by coupling the uncertainty of fire detection, alarm, and pre-movement with evacuation time.

The forecasting model is event-dependent and takes into account many social and environmental elements regarding different sorts of events, such as socio-economic situations and geographical features. This is due to the great range of emergency occurrences, including both natural and man-made ones. The business decision-making process in disaster operations management varies greatly depending on the type of occurrence, taking into account factors like severity, impacted region, population density, and local environment, among others.

There are many different types of hazards present worldwide. The health of vulnerable people is placed at risk by natural, biological, technological, and sociological dangers, which also have the potential to seriously impair public health. For instance, the authorities in charge of providing clean water are responsible for the prevention of waterborne illnesses, while law enforcement and road transportation agencies are in charge of reducing traffic accidents. Zoonotic illnesses (diseases spread from animals to people) need coordinated action from the agricultural, environmental, and health sectors. These increases in new or reemerging diseases are attributed to a number of factors, including global warming, low vaccination rates in high-risk and vulnerable populations, growing vaccine resistance and skepticism, rising antimicrobial resistance, and expanding coverage, frequency, and speed of international air travel.

A professional who develops plans for emergencies, accidents, and other calamities is known as an emergency management director.

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https://doi.org/10.1016/j.dajour.2023.100320
Received 29 June 2023; Received in revised form 11 August 2023; Accepted 5 September 2023
Available online 9 September 2023
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Directors of emergency management work together with the leadership team of an organization to evaluate possible hazards and create best practices for handling them. Designing emergency procedures and developing preventative actions to lessen the risk of emergency circumstances occurring fall under their purview. Directors of emergency management play a crucial part in ensuring the safety of all employees and equipping staff to act effectively in case of an emergency.

Plans for disaster preparation choose appropriate organizational resources, lay down the tasks and roles, establish rules and processes, and plan exercises to increase preparedness for disasters. The effectiveness of the response activities is improved when the needs of populations affected by catastrophes are anticipated. The effectiveness of the response operations is increased by increasing the ability of workers, volunteers, and disaster management teams to deal with crises. Plans could consist of the following: Sites for temporary refuge, and routes for evacuation water and energy sources for emergencies. Additionally, they might talk about stockpile requirements, communication protocols, training plans, chain of command, and training programs.

One of the most crucial metrics for gauging the effectiveness of an evacuation is the time it takes. Residents who are detained for an extended period of time represent a serious threat to staff safety because of the unpredictability of events. A building's inhabitants who attempt to flee during a fire accident exhibit a range of response times (RTs) between the time they are given a warning and the decision to leave. A number of complex factors, such as occupants' familiarity with evacuation routes, their ability to operate evacuation amenities and fire protection apparatuses, the number of people in the area, and occupants' psychological and physical conditions and behaviors, can affect how affected personnel are evacuated from a disaster site. Different factors have an impact on evacuation time (ET).

The results indicate that it is a variable influenced by a significant number of uncertain factors, including emergency evolution dynamics, human behavior under emergency conditions, and the environment. Author [3] demonstrated the benefits of developing appropriate emergency response plans using safety and industrial hygiene resources to mitigate or prevent harm to factory personnel and nearby community residents caused by chlorine gas leaks.

Everyone on your team has to be knowledgeable on how to spot leaks and react to them in order to keep your employees safe when handling chlorine. Since chlorine has a strong, unpleasant scent that resembles that of a potent cleaning solution like bleach, most chlorine leaks are quite easy to detect. Every facility that works with chlorine has to have an emergency kit on hand. This kit should include a variety of tools that may be used to stop or limit leaks around plugs, valves, or the side wall of a tank or cylinder used to store chlorine. Breathe in some fresh air and leave the location where the chlorine gas was emitted. If your community has an emergency notification system, be sure you are familiar with it. For directions, consult local authorities and emergency bulletins. If the chlorine discharge occurred outside, seek protection inside. To ensure that the contamination does not enter, make sure all windows are closed and ventilation systems are off. Leave the location where the chlorine was discharged if you are unable to get inside. Get outside and look for higher ground if the chlorine discharge occurred indoors. Open the windows and doors to the outdoors if the chlorine leak was caused by chemicals or home cleaners to allow in fresh air. Wait until the gas has dispersed before leaving.

Instead of being a final product or result, emergency preparedness should be a process. Its most basic requirement is to match resources with urgent demands in a timely manner to prevent procrastination and delays. Good emergency plans are both practical and realistic. Higher levels of reaction must be called for if the emergency's scope surpasses or overwhelms local resources. The difficulty of handling catastrophes successfully in a globalizing world that is vulnerable to fast physical, social, and economic change depends on excellent planning and foresight, as well as the capacity to combine disjointed aspects of the emergency response into cohesive plans.

Cloud service composition (CSC) has created new opportunities for vendors and consumers by providing sophisticated and composited services to meet user needs in dynamic and ambiguous contexts such as the cloud. Petri net (PN)modeling methods We focus on agent-based problem-solving strategies with business decision-making capabilities for CSC, which are based onMulti-criteria business decision-making methods (MCDM) methods for dealing with automated selection in CSC and PN techniques for modeling such context [4]. Petri nets are used as modeling tools in the discrete-event dynamic process known as the multi-agent system. In comparison to alternating current microgrids, direct current microgrids stand out for their ease of control and power management. They also offer a number of benefits, including higher conversion and transmission efficiency, greater reliability even in remote locations, convenient control, lower costs, and less filter effort due to the absence of reactive power, phase synchronization, high inrush current, etc. [5].

A rational actor must interact if enhancing subjective utility necessitates interaction with other agents. If there is contact between rational agents, at least one of the agents is trying to maximize his utility. Agents collaborate if their aims are the same. If their aims conflict, they engage in competition. The majority of these interactions occur between these two extremes. An interacting agent would do well to predict the objectives of other agents. A more well-informed actor may foresee some aspects of how other agents will act in response to their objectives [6]. In these situations, strategic thinking is required. A contact in which strategic thinking occurs is referred to as a strategic interaction (SI). In game theory, SI or games are examined. The game theory takes into account reason and the potential to forecast rational behavior. The existence of widespread awareness of reason is assumed. This implies that each participant in an interaction believes in the reason of the others and that they, in turn, believe in his rationality, and so on [5].

The equilibrium is the expected behavior of players or participants in an interaction. If one of the players strays from equilibrium, nobody wins. Because of this, it is termed equilibrium. In finite games, there is at least one equilibrium [7]. At least two application agent and mechanism designs are required for artificial intelligence games. We have a game in agent design and must calculate appropriate behavior. We have an expectation about the behavior and must develop game rules in mechanism design. These two goals can be addressed theoretically by running algorithms over a game tree, or practically by creating an environment in which various real players can interact. Most games are written in low-level programming [8]. Game rules are more easily editable. Algorithms may be created that change game representation in every way imaginable, such as 'reduce number of players' or ' remove simultaneous turns'. Game representations may also be used to create evolutionary mechanisms [9].

Logical Petri nets can further simplify the network structure of real-time system models, making it easier for us to analyze the properties of the system at a conceptual level, while also alleviating the problem of state space explosion to some extent. Petri nets can not only characterize the structure of a system but also describe its dynamic behavior. Currently, many scholars have proposed extended forms of Petri nets, such as logical Petri nets [10], timed Petri nets [11,12], and colored Petri nets [13,14], and their applications are becoming increasingly widespread. Multi-agent games involve multiple elements, such as players, strategies, utilities, and information equilibrium. The existing modeling elements of logical Petri nets cannot accurately describe these elements, so improvements need to be made to logical Petri nets. Based on the existing modeling elements of logical Petri nets, modifications or additions of new modeling elements are needed to model game elements, enabling the new model to accurately describe dynamic game problems in multi-agent systems.

We consider a mean-field game (MFG)-like scenario where a large number of agents must select between a set of various potential target destinations. This scenario is inspired by effective biological collective

decision mechanisms such as the collective navigation of fish schools and honey bees searching for a new colony. The mean trajectory of all agents represents how each person impacts and is impacted by the group's choice. The model can be seen as a stylized representation of opinion crystallization in a political campaign, for instance. The initial spatial position of the agents determines their biases initially, and then in a later generalization of the model, a combination of starting position and a priori individual preference. The existence criteria for the specified fixed point-based finite population equilibrium conditions are developed. In general, there may be several equilibria, and for the agents to compute them properly, they need to be aware of all the beginning circumstances.

Game theory is an area of applied mathematics that offers methods for examining scenarios where parties, or players, make interdependent decisions. Each player must consider the potential decisions or strategies of the other players because of their interdependence. A game's solution outlines the best choices made by the participants, who may have conflicting, overlapping, or mixed interests, as well as the possible results of those choices.

According to the game theory, various players choose their strategies based on the results they may expect from them. A pure strategy is one that gives players the greatest benefit or the best result. As a result, it is seen to be the optimal approach for every gamer. The optimal strategy for an organization (or player) is one that is unaffected by the strategies of other organizations (or players). A maximin strategy is one in which a player or organization maximizes the possibility of making the least amount of profit in order to lower the level of risk. When using a minimax strategy, a player's primary goal is to minimize loss and maximize gain. It is an instance of a mixed tactic. A gamer can therefore use various strategies.

The number assigned to a certain bundle by the utility function is known as utility, as is the satisfaction a customer receives from a specific bundle. In absolute terms, the utility value for each bundle means nothing; there is no consistent scale against which we may assess contentment. Its sole objective is to decide which bundles are preferred above others in terms of usefulness. As a result, utility functions rank consumer preferences by allocating a number to each bundle.

Both the idea of a rational solution and the dominant-strategy equilibrium are well-founded. We would anticipate that players will play in accordance with the dominant-strategy equilibrium whenever such an equilibrium occurs if they are rational and cautious in the sense that they assign positive probabilities to each of the other players' plans. On the other side, rationalizability identifies precisely what the game's definition (i.e., accepted principles of rationality) implies. The participants must all be employing rational strategies if it is assumed that they are rational (that is, they maximize the anticipated value of their utility function). Additionally, every rational strategy is rationalizable in the sense that a player can employ it while still holding the belief that rationality among players is a given.

Business decision-making is an essential stage in a wide range of tasks, and those in charge of it must be able to discover appropriate answers under all conditions. Some decisional systems have been designed to assist people by attempting to give intelligent behavior. Logical inference is one of the most useful artificial intelligence processes for business decision-making [15]. Logical game choice The dynamic game process is analyzed using a Petri net and the reachability graph generated by reachability identification. The technique for generating the reachability graph is presented, as well as how to solve the dynamic game choice issue, resulting in the ideal emergency pre-arranged plan, and the resource conflict in the emergency process is analyzed by the logic game decision model of emergency [16]. Stochastic decision Petri nets (SDPNs) are a type of stochastic Petri net with incentives and a control mechanism based on the deactivation of controlled transitions. When such nets are transformed into Markov decision processes (MDPs), the number of states might possibly explode owing to concurrency. As a result, we limit ourselves to cases in which nets are either SAFC nets or occurrence nets, and policies are specified by a consistent deactivation pattern. Through a tight relationship to Bayesian networks, we establish complexity-theoretic findings for such cases [17].

When disaster strikes, it is crucial to act quickly to limit damage to people and their possessions and stabilize the situation. These measures are structured within an emergency response procedure, as detailed in a disaster preparedness plan. However, it is now of the utmost importance to develop reliable emergency plans well before any potential crises. The emergency evolution dynamics, human behavior, and the environment all make it difficult to predict when it will be safe to evacuate and how long it will take. As a result, better emergency response plans need to be developed to address these unknowns and boost overall readiness and response capacities.

Finding dangers specific to your organization, geography or area, market, goods, and procedures is a good place to start. Once risks have been identified, evaluate them for likelihood and predicted impact severity. Next, note the indicators of each crisis. These evaluations will help prioritize tasks and allocate resources. It is crucial to regularly conduct exercises or simulations to test the plan in order to swiftly implement a Crisis Management Plan (CMP) during a crisis. Drills or practices simulating various crisis circumstances may highlight an organization's readiness weaknesses as well as its strong points.

The first stage in developing an emergency response plan is to undertake a detailed threat assessment to identify and analyze the sorts of events that could impact your organization. Specific dangers differ by geography, industry, and firm, and your mitigation tactics will differ accordingly. Even when lives are not immediately at risk during a crisis, rapid communication is critical. Employees will seek to their leaders for reassurance and advice in the event of an emergency. These individuals should be in charge of implementing your emergency response plan, addressing critical questions, and ordering an evacuation if necessary. There are several factors to consider while allocating responsibilities. You must ensure that your response team is present, dependable, and capable of acting promptly in the event of an emergency.

The primary aim of this study is to improve emergency response plan creation by utilizing sophisticated modeling approaches. Using stochastic modeling, we first analyze the potential influences of variables, including fire detection, warning systems, and pre-movement behaviors on evacuation timeframes. This investigation aims to shed light on the interplay between probabilities and evacuation times. Second, this study intends to show how useful it is to draw on safety and industrial hygiene resources when crafting emergency response plans. These strategies can ensure the safety of employees and the surrounding community by reducing or eliminating the adverse effects of catastrophes. By accomplishing these goals, this study will contribute to emergency management by providing helpful information and methods for enhancing emergency preparedness and response operations.

Camera-equipped IoT/Edge devices may be placed strategically along slopes, ridges, and high-altitude regions, automatically looking out for signs of smoke or fire. An alert might be set off by each individual sensor, or you could send the sensor data to a central hub that gathers, processes and analyses the sensor data to determine the likelihood of a house fire. It is occasionally possible to reevaluate the level of structural fire protection necessary on escape routes by installing an efficient, dependable automated fire detection system that is coupled with an efficient fire warning system. This may offer a more practical and affordable fire safety measure. Pre-movement periods have been hypothesized to be predictable despite their complexity based on observational evidence that shows that their mean and variance may be approximated for certain settings. Researchers have created theoretical models based on decision-theory or have computed pre-movement durations using empirically established distributions.

The main contributions of this paper are as follows: (1) Combining sequential game models with Petri nets, we propose the definition of logical game decision Petri nets and introduce optimal decision transitions. Optimal decision transitions compare utility values represented

by tokens to determine the optimal decision solution. (2) Based on the logical game decision Petri net model, we model and analyze the dynamic game process of sudden events and construct reach ability graphs to provide reachability sequence identification. (3) Using the logical game decision Petri net model, in combination with a comparative analysis of case studies, we implement the optimal solution and depict the state transition in sudden event scenarios.

2. Fundamentals

Definition 1. M = (Q, U, G) is a net [18–22], where:
(1) Q is a finite set of places.
(2) U is a finite set of transitions, Q ∪ U ‘ ∅ and Q ∩ U = ∅;
(3)G Þ (Q × U) ∪ (U × Q) is a set of directed arcs.

Definition 2 (*Input/Output Sets*)**.** Let y∈ Q ∪ U be any element of net M. y’ = {z ð (z, (y) ∈G} is the input set or pre-set of y, and y* = {z ð (y, (x) ∈ G} is the output set or post-set of y. M = (Q, U, G) is a pure net that satisfies ∀ uthe ∈U: u* ∩ u’ = ∅.

Definition 3. A quadruple = (Q, U, G, N) is called a Petri net if and only if:
(1) M = (Q, U, G) is a pure net.
(2) N: Q ™ M is an identification function, where N_0 is the initial marking.
(3) has the following transition firing rule:
(1) For a transition u∈U, if ∀q∈u′: N(q) g 1, then transition t is enabled under marking N, denoted as N[uë.
(2) If N[uë, then transition u is enabled under marking N and generates a new marking N′, denoted as N[uëN′ ð.

Definition 4. Let LPN = (Q, U, G, I, P), LPN = (LN,N) be a remote Petri net if and only if:

$$N'(s) = \begin{cases} N(q-1); & \text{if } q \in -u^* \\ N(q+1); & \text{if } q \in t - u \\ N(q); & \text{other} \end{cases}$$

(1) Q is a finite set of places.
(2) U = U_D∪U_I∪U_O is a finite set of transitions, U∪Q ‘ ∅ , ∀u∈ U_D∪U_I∪U_O:
In:
(1) U_D represents the transition of classic Petri net;
(2) U_I represents the logical input transition set of transition U and ∀u∈U_I; the rational input expression f_I restricts all the former locations of U_I;
(3) U_O represents the logical output transition set of transition U, and ∀u∈U_O, the analytical output expression f_O restricts all subsequent stages of U_O.
(3) G Þ (Q × U) ∪ (U × Q) is a finite set of arcs.
(4) I am a logic-restricted input function; for ∀ u∈ U_I, I(u) = f_I is a logic input expression.
(5) O is a logic-restricted output function; for ∀ u∈ U_O, O(u) = f_O is a logic output expression.
(6) N: Q™{0,1} is the identity function, ∀ q∈Q, N (q) represents the number of tokens contained in place q.
(7) The transition triggering rules are:
(1) For ∀u ∈ U_D, if ∀u ∈ U_D, the triggering rule of transition t satisfies the rule of transition triggering in Definition 3;
(2) For ∀u ∈U_I, I(u) = f_I, if f_IðN =. When triggered under the symbol N, it evolves into a new N′: ∀u ∈u: N′ (q) = N (q); u*; N′(q) = N(q) + 1; ∀qtu*u ∩ u* : N′(q) = N(q)
(3) For ∀u ∈U_O, O(u) = f_O, if ∀q∈.u, N(q) = 1, then t is enabled under N; if t is enabled, it can be triggered, and t is After triggering under the symbol N, it evolves to a new symbol N′: ∀ q∈ q) for t, it should satisfy f_OðN′ =. U. That is, t must satisfy the logical expression f_O in N′.

3. Introduction to the sequential game process of emergency decision-making

In emergency decision-making, game theory is integrated into Petri Net. Its main features are:
(1) The state of emergency develops and changes dynamically.
(2) The information of emergencies is constantly improving, from fuzzy to clear, and from incomplete to complete; (3) The response plan formulated under the condition of incomplete information is easy to adjust in time when the information is complete; Based on the state of the event, the staged disaster relief results, and the evolution of the emergency state, a new response plan is generated [23–26]. That is, the sequential decision game model can adjust the strategy according to the state of the emergency.
(1) People in the game (emergencies and emergency decision makers)
Crisis events are one side of the game, and emergency decision-makers are the other side of the game.
(2) Strategy space
This paper gives a mathematical description of the crisis events of both sides of the game and how the emergency decision-maker generates the emergency plan. The state space of the crisis event is: {K_1, K_2, K_3, ., K_n}. The strategic combination of the emergency decision-maker is the configuration and scheduling of emergency resources in the process [27–29].
(3) Profit Function
We assume that in the emergency business decision-making process of a crisis event, the game model between the “decision maker” and the “emergency event” is a dynamic zero-sum game; that is, the loss of one party in the game is the gain of the other party.
(4) Game Goal
The emergency rescue process aims to supply resources in the disaster area while minimizing the cost of emergency resource consumption. The objective function is:

$$f\ V_i = \min \sum_i V_i \quad (1)$$

The dynamic game process of crisis events in the emergency business decision-making process is as follows:
In **Step**1 h_0 stage, a crisis event first breaks out, chooses its own state K_i and causes a specific loss, marked as the initial state information.
In the **Step**2 h_1 stage, the crisis event decision-maker judges the probability of its state according to the observed state information of the crisis event. Then it selects the optimal resource allocation and scheduling plan according to the principle of maximizing the expected utility to achieve better rescue. Effect
In **Steps** 3 and h_2, the emergency decision-maker evaluates the staged rescue situation. The emergency may evolve from state K_i to a new state K_j. The state information is corrected or inferred, and a unique optimal solution is selected for rescue according to the principle of expected utility maximization.
Step 4 Implementing the new rescue plan makes the state of the emergency evolve into another new state. The emergency decision-maker needs to formulate a new strategy based on the obtained information and implement a decision for the rescue process. Repeat in turn until the crisis event is effectively resolved Control or release.

4. Logic game decision Petri net

Definition 5 (*Logical Game Decision Petri Net, LGDPN*)**.** A logical game decision Petri net can be represented by a tuple, LGDPN = (M, Q, U, G, , , D.u, ', , T, N).
(1) M is the set of rational players, M = {m_1, m_2, …, m_i}, where ð Mð = 2, indicating the participants in the game considered in this paper.
(2) Q is a finite set of places, representing the set of states in which rational players can take actions, Q = Q_A∪Q_D.
Q_A represents ordinary places in a simple Petri net.

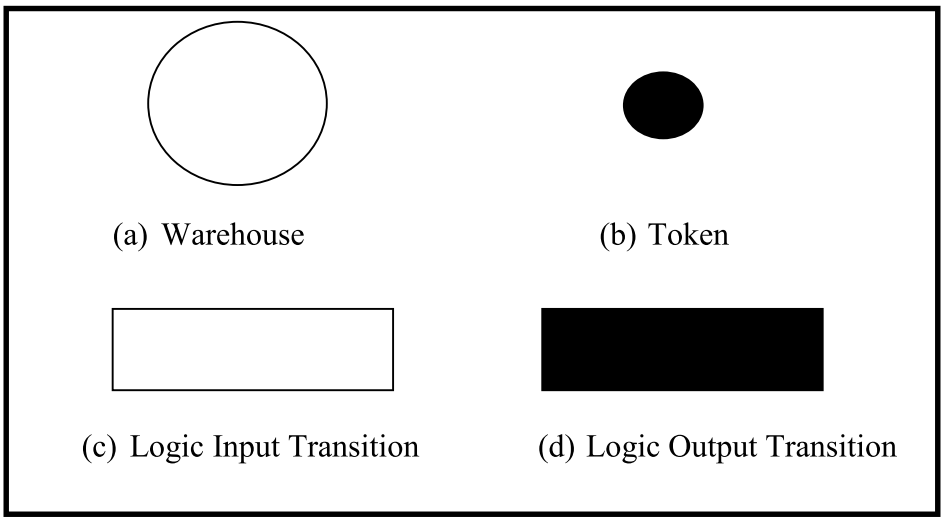

Fig. 1. Element representation in the logic game decision Petri net model.

Q_D represents dynamic probability places.
(3) U is a finite set of transitions, representing the set of actions that rational players can take, $U = \{u_1, u_2, \ldots, u_n\}$, $U = U_R \cup U_L \cup U_K$.
(1) U_R represents transitions in a simple Petri net, with firing rules similar to a simple Petri net.
(2) U_L represents logical transitions, $U_L = U_I \cup U_O \cup U_{IO}$, where U_I represents logical input transitions, U_O represents logical output transitions, and U_{IO} represents logical input–output transitions.
(4) G is a finite set of arcs, G Þ (Q × U) ∪ (U × Q).
(5) is a set of probability vectors, $= \{b_1, b_2, \ldots, b_n\}$, where the $b_i \in [0,1]$, represents the probabilities before state transitions of event places.
(6) is a set of probability vectors, $= \{c_1, c_2, \ldots, c_n\}$, where $c_i \in [0, 1]$, representing the probabilities of transitions during state transitions.
(7) D.u represents the guarantee rate and cost of tokens, indicating the cost attribute and guarantee rate of tokens themselves, representing the value of the utility function.
(8) ' is a finite decision set of rational players, ∀q∈Q, ∀u∈U, 'êq, uë represents the decision of rational player n to choose action u from state q.
(9) is a finite set of resources, Resource = $\{s_1, s_2, s_3\}$, $(s_i) = 0$ indicates a reusable resource, and $(s_i) = 1$ indicates a non-reusable resource.
(10) V is the utility function of rational players, defined on the finite set of transitions U.
(11) N is the identity set of the network, ∀ q ∈ Q, $N_i(q)$ means that in the current identity all tokens in place q under N_i.

Definition 6 (*Utility Function V*)**.**In a logical game decision Petri net, $V_r^p(u) = V('(q;uë); V_r^p(u); V U_n^p(u)$ represents the fitness function that occurs when rational player n is in state p and takes action u. $V('êq;uë))$ represents the utility function of rational player n when choosing decision 'êq;uë in state q. The utility function is represented by a two-dimensional vector, $V('êq;uë) = [\ ;d]$; where represents the guarantee rate and d represents the cost.

The guarantee rate is determined by the guarantee function, = G(S, K, S, O,Q), where represents the guaranteed rate, G represents the guarantee function, R represents the resource situation, K represents the state of the emergency event, S represents the decision maker's choice of action, O represents the decision maker's objective, and Q represents the event state.

Definition 7. The state transition probability function $q_{ij} = f\ s_i; i; j$. In the logical game decision Petri net, the state transition probability function records the probability of the token in place q_i transitioning to place Q_j after choosing the decision transition u_i. Here, $_i$ represents the guaranteed rate after the implementation of the rescue plan, irepresent the state chosen in the first stage of the game for the crisis event, and j represents the new state evolved in the second stage of the crisis event. The matrix representing the probability of transition between crisis event states can be expressed as an

$$\text{downright}\begin{matrix} `q_{11} & 5 & q_1 & a \\ r & & & s \\ r\ 4 & 7 & 4 & s \\ r & & & s \\ pq_{n1} & 5 & q_{nn} & q \end{matrix}$$

Definition 8. Dynamic probability places Q_D are used to record the probability changes in the state transitions of players in the game. For $\forall q_i \in Q_D; u \in q_i^*; 0 < q_i; q < 1$ is one of the conditions for the transition to be triggered.

Definition 9 (*Optimal Decision Transition* U_k)**.** The decision transition U_k is triggered when the optimal solution is selected for the crisis event after dynamic gaming.

Definition 10 (*Sequential Game*)**.** In a probability game Petri net, the players in the game N adopt a sequential order when taking action strategies Q, which is called a sequential game.

The execution mechanism of Petri nets can be used to model state transition systems, for example, when the occurrence of an event leads to the occurrence of other events. Typically, the state of a system (occurrence of events) can be represented by places. State transitions can be represented by transitions in the Petri net, as shown in Fig. 1.

Fig. 2 presents the work routing structure of the logical game decision Petri net model. Fig. 2(a) and (b) depict the sequential structure and concurrent structure, respectively. If N [uë, the decision transition in the U_K is triggered in the representation of N to pass token values to successor places. Fig. 2(c) represents the selection structure where the tasks responded by decision transitions u_1 and u_2 are different, resulting in different values of utility function transmitted by tokens.

Assuming that the emergency rescue is in a certain stage, the probabilities of various states being true are $u_1 \mid g_k\ ;\ u_2 \mid g_k\ ; \ldots ;$ $u_j \mid g_k$. The decision cost of adopting the rescue plan $_i$ at this stage is given by [30–34]:

$$G\ g_k;\ _i = \sum_{j=1}^{\text{Ë}} u_j \mid g_k\ D_{ij} \tag{2}$$

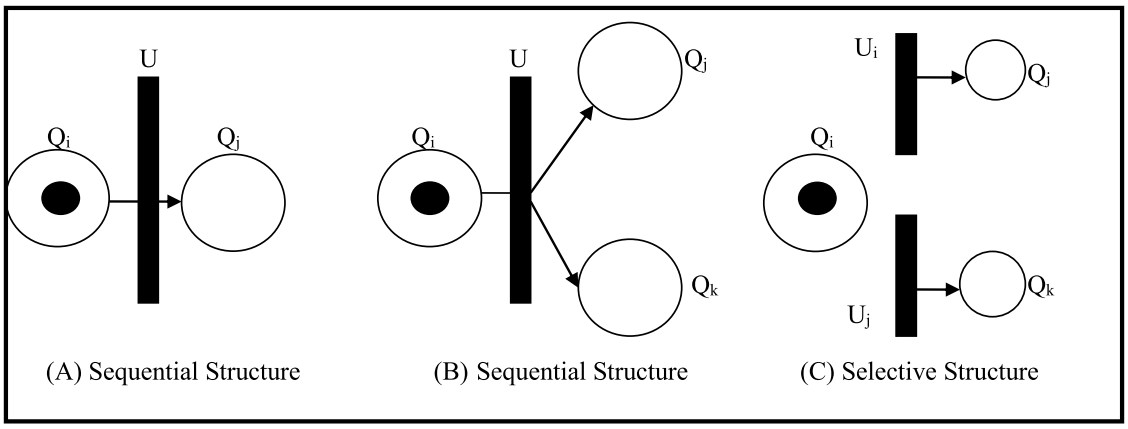

Fig. 2. Structure diagram of the logical game decision Petri net.

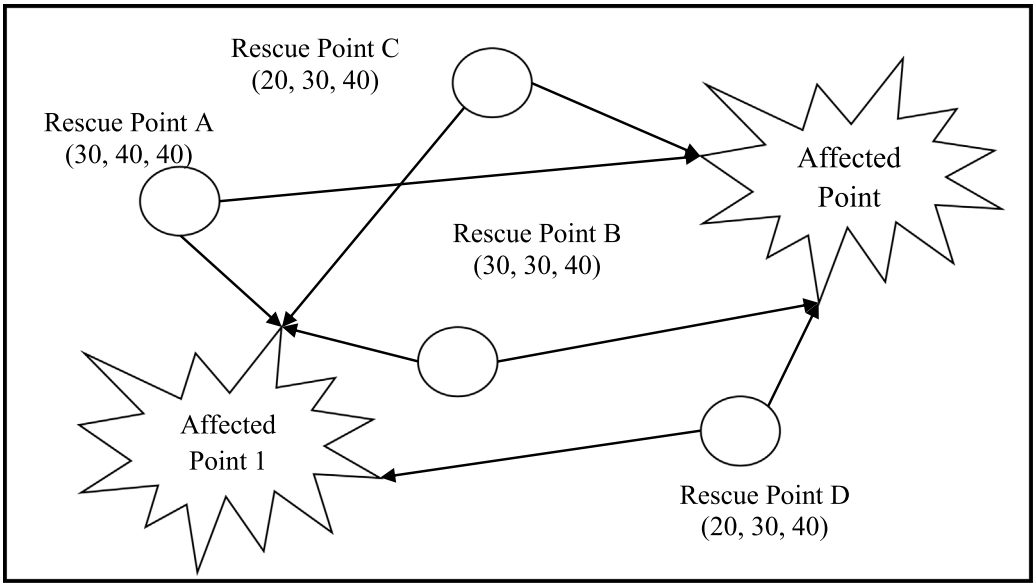

Fig. 3. Distribution topology of rescue points and disaster points.

The emergency decision-maker selects the plan $_i^*$ with the minimum decision cost within the range of collected information, as follows:

$$G\ g_k;\ _j = \min\ G\ g_k;\ _1 ;G\ g_k;\ _2 ;\ldots;G\ g_k;\ _j \tag{3}$$

Analyzing Emergency Decision Process Using Logical Game Decision Petri Net

4.1. Model assumptions

Taking an oil pipeline accident in a certain region as an example, the initial event of the pipeline network accident is an oil leak caused by pipeline rupture [35–37]. Subsequently, other events such as fire, explosion, and contamination may occur. Among them, there is one main disaster area, while the rest of the areas are not considered for now. The disaster area requires three emergency resources (medical supplies, firefighting equipment, and rescue devices), which are provided by four nearby rescue points. The distribution topology of the disaster area and rescue points is shown in Fig. 3. Considering the resource loss rate and transportation cost, it is assumed that by providing sufficient resources to the disaster area within a certain time during the emergency event handling process, the emergency event can be completely controlled. If the resources transported to the disaster area are insufficient, the emergency event can only be partially controlled.

4.2. Model description based on logical game decision

Petri Net Logical Petri nets are suitable for describing and analyzing systems with prioritized batch processing functions [38–42]. Therefore, it is more appropriate to use the logical game decision Petri net to model the dynamic game decision behavior of emergency events. Based on the description of the emergency event process in Section 2 and the rules of logical Petri nets, a dynamic game decision model for emergency events based on the logical game decision Petri net is established, as shown in Fig. 4. The meanings of places and transitions in the dynamic game process LGDPN model of the emergency event are listed in Tables 1 and 2.

4.3. System reachability graph

The system reach ability graph [43–45] can visually depict the state transitions of emergency events and the game situations with emergency decision-makers. The reachability graph is obtained based on Algorithms 1 and 2. In the reach ability graph, $V_1^{K_1}; V_2^{K_1}; V U_3^{K_1}$ respectively represent the utility function values of rescuing through game stage selection methods $_1$, $_2$, $_3$ when the emergency event starts in state $V_1^{K_1}; V_2^{K_1}; V U_3^{K_1}$ respectively represent the utility function values of rescuing through game stage selection methods $_1$, $_2$, $_3$ when the emergency event starts in state K_2; when the emergency event

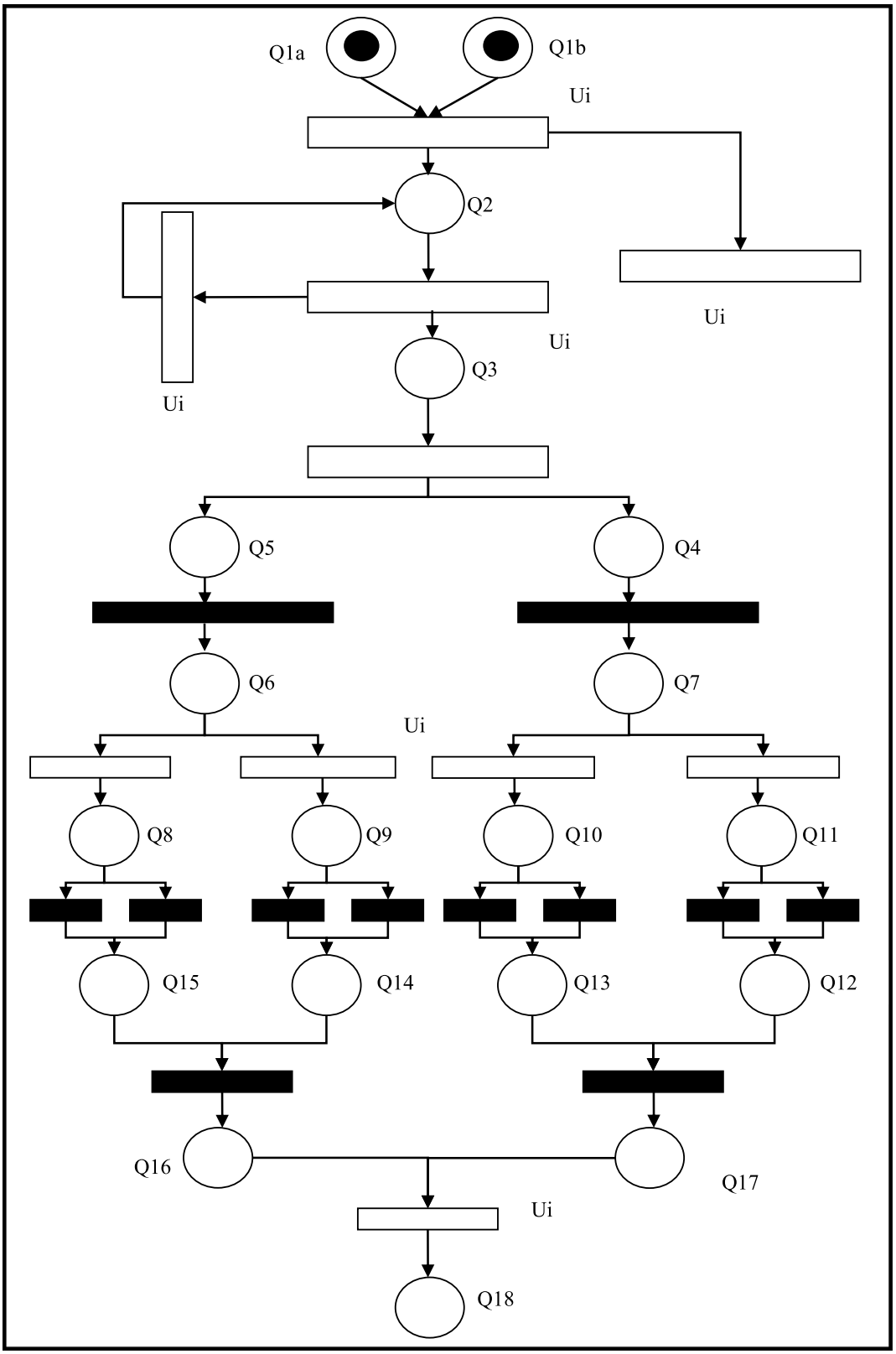

Fig. 4. Dynamic game process LGDPN model of emergency events.

is in state K_1, the utility values of selecting rescue plans $_1$, $_2$ in the first stage and rescue plans $_1$, $_3$ in the second stage are $V^{K_{11}}_{12}; V^{K_{12}}_{13}$ when the emergency event is in state K_2, the utility values of selecting rescue plans $_1$, $_2$ in the first stage and rescue plans $_1$, $_3$ in the second stage are $V^{K_{21}}_{12}; V^{K_{22}}_{13}$. The solution spaces are $T_1 = \ _1 \ ; T_2 = \ _2; \ _3$.

Algorithm 1. Trigger Optimal Decision Transition T^*_K

Input: LGDPN = (M, Q, U, G, , , D, t, ', , V, N) Output:

Trigger situation of the optimal decision transition

Table 1
Place information.

Place symbol	Place symbol meaning
Q_{1a}	Incident 1
Q_{1b}	Emergency 2
Q_2	Emergency Management Center
Q_3	Corresponding level status of crisis events
Q_4	Alarm ended
Q_5	Emergency status K1
Q_6	Emergency status K2
Q_{7-8}	Event status information
$Q_{9;11}$	Event state evolution K1
$Q_{10;12}$	event state evolves to K2
Q_{13-16}	Event status information after rescue
Q_{17-18}	The emergency rescue benefit information
Q_{19}	Emergency business decision-making discussion information
Q_{20}	Program Determination Information
Q_{21}	Decision end information

Table 2
Transition Information.

Symbol of change	Transition symbol definition	Resource requirements
u_1	Alarm	–
u_2	Determine Response Level Status	–
u_3	Accident is minor	–
u_4	Do not deal with the accident	–
u_5	Emergency Response Classification	–
u_{6-7}	Decision Response Method 1	R(20,30,30)
u_{8-11}	The event state evolves with a certain probability	
u_{12-15}	Decision Response Method 2	R(10,10,20)
u_{16-19}	Decision Response Method 3	R(30,30,60)
u_{20-21}	Comparison of program business decision-making benefits	–
u_{22}	Discussion on business decision-making options	–
u_{23}	Optimal decision determination	–
u_{24}	End of emergency decision	–

1. initialize utility function value V } 0, and the emergency event appears in state K_i.
2. IF ∀u ∈ U_K; U_K ' ∅; N(q) > 1 THEN
   - **2.1** Check the place status of U_K, and perform phased rescue plans.
   - **2.2** Calculate the utility value V_m '$_{Ki}$êq; uë when different plans are adopted in the game process.
3. IF U_K.(K_i) state is normal, THEN compare the utility values of different plans V
   - **3.1** IF U '$_i$êp; të > U '$_j$êp; të
   - **3.2** U '$_i$êp; të } U '$_j$êp; të
   - **3.3** Trigger the optimal decision transition U_K^*, and output the optimal utility value V and decision transition u ELSE
   - **3.4** $U_K{}^*$ is not triggered.

Algorithm 2. Reachability Graph Algorithm
Input: LGDPN = (M, Q, U, G, , D, t, ', , V, N)
Output: Reach ability Graph RG(LGDPN)

1. Set N_0 as the root node of RG(LGDPN) and label it as new(N), and let N set be the set of reachable markings.
2. Select any node with the new(N) label and denote it as N.
3. If a node on the directed path from N_0 to N with the same label as N exists, THEN change the label of N to old(N) and go back to step 2.
4. For ∀u ∈ U such that N uëN′.
   - **4.1** IF N′ is equal to any label N" in the set of markings N set, THEN draw a transition arc from N to N" labeled as u, label N as old(N).
   - **4.2** IF N′ is not equal to any label in the set of markings N set, THEN label N′ as new(N), label N as old(N), draw a transition arc from N to N′ labeled as u, and add N′ to the set, and go back to step 1.
   - **4.3** IF ¬N[u, THEN label N as a terminal node (TerminalNode), go back to step 2, until there is no marking labeled as new(N) in the set N set.

Table 3
Emergency event triggered to an emergency center.

Status	Value
N_0	[1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0]
N_1	[0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0]
N_2	[0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0]

Table 4
Emergency business decision-making process when the state of the emergency event is $K_1 - K_1$.

Status	Value
N_3	[0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0]
N_4	[0,0,0,0,0,0,0,(1, $V^{K_1}_1$),0,0,0,0,0,0,0,0,0,0,0,0,0]
N_5	[0,0,0,0,0,0,0,0,0,(1, $V^{K_1}_1$),0,0,0,0,0,0,0,0,0,0,0,0]
N_6	[0,0,0,0,0,0,0,0,0,0,0,0,0,(1, $V^{K_1}_1$, $V^{K_1}_2$),0,0,0,0,0,0,0]
N_7	[0,0,0,0,0,0,0,0,0,0,0,0,0,(1, $V^{K_1}_1$, $V^{K_1}_3$),0,0,0,0,0,0,0]
N_8	[0,0,0,0,0,0,0,0,0,0,0,0,0,(1, $V^{K_{11}}_{12}$;$V^{K_{12}}_{13}$),0,0,0,0,0,0,0]
N_9	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0]
N_{10}	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0]
N_{11}	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1]

Table 5
Emergency business decision-making process when the state of the emergency event is $K_1 - K_2$.

Status	Value
N_{12}	[0,0,0,0,0,0,0,0,0,0,(1, $V^{K_1}_1$),0,0,0,0,0,0,0,0,0]
N_{13}	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,[1, $V^{K_1}_1$, $V^{K_1}_2$],0,0,0,0,0]
N_{14}	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,[1, $V^{K_1}_1$, $V^{K_1}_3$,0,0,0,0,0]]
N_{15}	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,[1, $V^{K_{11}}_{12}$;$V^{K_{12}}_{13}$,0,0]]
N_{16}	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0]
N_{17}	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0]
N_{18}	[0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0]

Table 6
Handling minor accidents when the state of the emergency event is K_1.

Status	Value
N_{19}	[0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0]
N_{20}	[0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0]

From Fig. 5, we can understand the emergency decision-making game process of the emergency event and the state transitions of the emergency event. The changes in markings during the state transition process are shown in Tables 3–8. Based on this analysis, we can determine how to generate reliable contingency plans at the minimum cost in the dynamic game process.

4.4. Case analysis

(1) Analysis of plan generation Assuming the relationship between resources R{r_1, r_2, r_3} and control rates are as shown in Figs. 6–8, determine the transportation situation of assumed resources based on the distribution of rescue points and disaster points, as shown in Table 9.

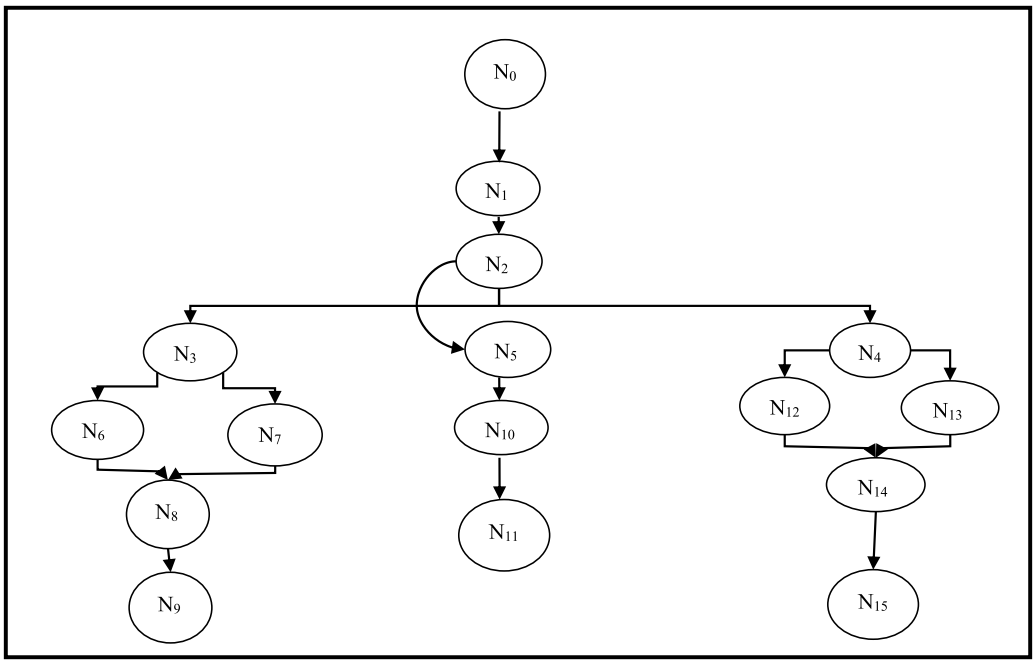

Fig. 5. Reach ability graph of the dynamic game process of the emergency event.

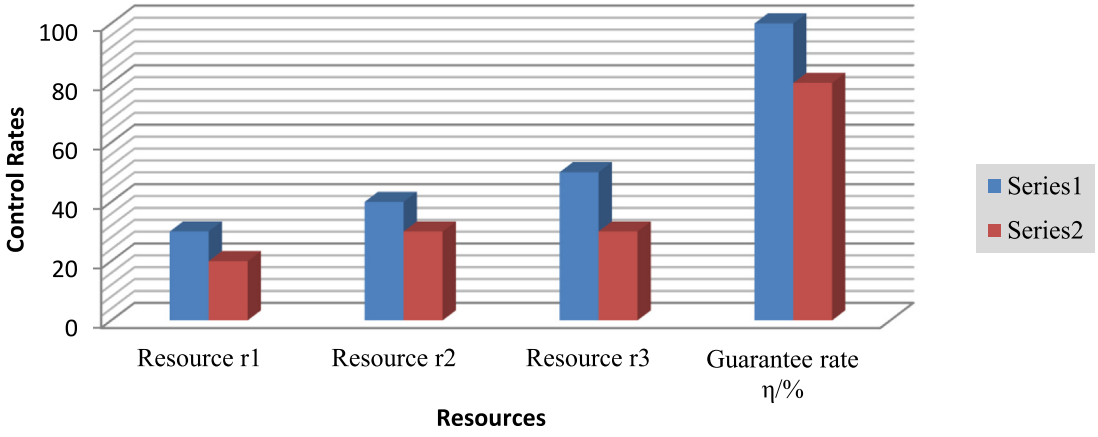

Fig. 6. Relationship between resources and control rates in crisis state K_1.

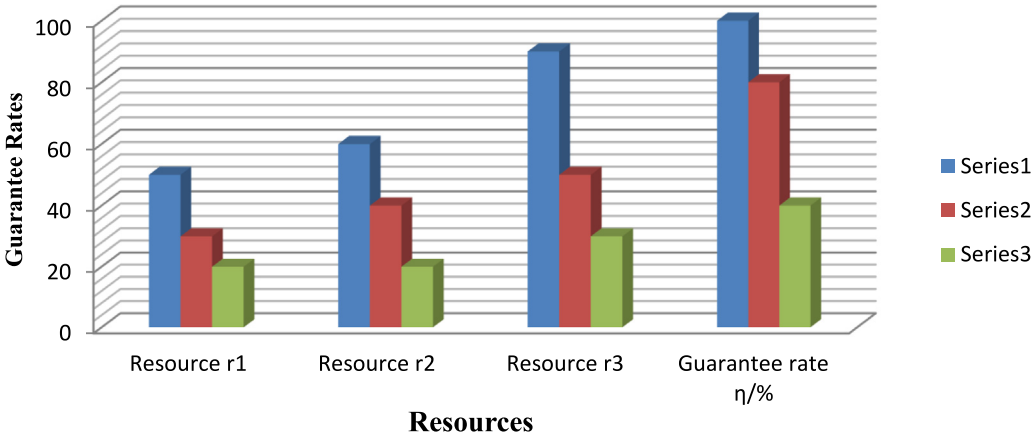

Fig. 7. Relationship between resources and guarantee rates in crisis state K_2.

Assuming that the emergency event and emergency management decision-makers engage in dynamic game behavior within a certain time frame, in this case, the game between the two parties is divided into two stages [46–48]. In the first stage h_1 of the game, the emergency